

12■Month Roadmap to Become a Data Engineer

This roadmap focuses ONLY on Data Engineering skills from beginner to advanced with free resources and projects.

Month 1 – SQL Foundations

Skills to Learn:

- Basic SQL queries, filtering, joins, aggregations, group by

Free Resources:

- SQLBolt Interactive Tutorial
- Mode SQL Tutorial

Project Idea:

- Practice SQL queries on sample datasets.

Month 2 – Advanced SQL

Skills to Learn:

- Window functions, CTEs, subqueries, query optimization

Free Resources:

- LeetCode SQL Problems
- BigQuery SQL Guide

Project Idea:

- Solve 40–60 SQL problems and analyze query performance.

Month 3 – Python Basics

Skills to Learn:

- Python syntax, data structures, file handling, scripting

Free Resources:

- freeCodeCamp Python Course
- Python Official Tutorial

Project Idea:

- Write scripts that read CSV files and transform data.

Month 4 – Python for Data Processing

Skills to Learn:

- Pandas, JSON/CSV processing, API data ingestion

Free Resources:

- Kaggle Pandas Course
- Pandas Documentation

Project Idea:

- Build a script that fetches API data and stores it in database.

Month 5 – Data Modeling

Skills to Learn:

- Star schema, snowflake schema, fact & dimension tables

Free Resources:

- Kimball Data Warehouse Concepts
- Snowflake Data Warehouse Guide

Project Idea:

- Design a data warehouse schema for an ecommerce dataset.

Month 6 – ETL Pipelines

Skills to Learn:

- ETL vs ELT, batch pipelines, transformations

Free Resources:

- Data Engineering Zoomcamp
- ETL vs ELT Explained

Project Idea:

- Build pipeline: extract CSV → transform → load into SQL warehouse.

Month 7 – Data Warehousing

Skills to Learn:

- OLTP vs OLAP, partitioning, analytics tables

Free Resources:

- BigQuery Documentation
- Data Warehouse Concepts

Project Idea:

- Create analytical tables for reporting dataset.

Month 8 – Big Data Concepts

Skills to Learn:

- Distributed systems, Hadoop ecosystem, MapReduce basics

Free Resources:

- Apache Hadoop Overview
- Spark Introduction

Project Idea:

- Understand distributed processing with sample datasets.

Month 9 – PySpark

Skills to Learn:

- Spark DataFrames, transformations, actions

Free Resources:

- Spark SQL Guide
- Databricks Spark Tutorials

Project Idea:

- Process large dataset using PySpark transformations.

Month 10 – Workflow Orchestration

Skills to Learn:

- Pipeline scheduling, DAGs, workflow automation

Free Resources:

- Apache Airflow Documentation
- Airflow Tutorial (Astronomer)

Project Idea:

- Automate ETL pipeline using Airflow scheduling.

Month 11 – Data Quality & Monitoring

Skills to Learn:

- Data validation, schema checks, pipeline testing

Free Resources:

- Great Expectations Docs
- Data Quality Guide

Project Idea:

- Add validation checks to ensure pipeline data integrity.

Month 12 – Capstone Data Engineering Project

Skills to Learn:

- Build full pipeline: ingestion → ETL → warehouse → analytics tables

Free Resources:

- Data Engineering Zoomcamp Projects
- Data Engineering Project Tutorial

Project Idea:

- Create end-to-end pipeline and publish on GitHub with architecture diagram.